

Topological Group.

Def. Let G be a Group under the binary operation $\cdot: G \times G \rightarrow G$.

A Topological Space $(G, \mathcal{L})$ satisfying T_1 condition, and

$G \times G \rightarrow G: (x, y) \mapsto xy$ and $G \rightarrow G: x \mapsto x^{-1}$ are continuous

is called the Topological Group.

Lemma. Let G be a group and T_1 -space. Then

G is Topological Group if and only if $G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$ is continuous.

Proof. Suppose that G is Topological Group. Then

$\varphi(x, y) = xy$ and $\psi(y) = y^{-1}$ are continuous, therefore

$(x, y) \mapsto xy^{-1} = \varphi(x, \psi(y))$ is continuous, being composition of continuous maps

Conversely, suppose that $\mu: G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$ is continuous.

Then, for fixed $x = 1$, $\mu(1, y) = y^{-1}$ is continuous, so $x \mapsto x^{-1}$ is continuous.

Now, $y \mapsto y^{-1}$ is continuous, so

$(x, y) \mapsto x \cdot (y^{-1})^{-1} = xy$ is continuous, being composition of continuous.

□

$GL_n(\mathbb{R})$.

Consider $GL_n(\mathbb{R})$ as the subset of $\mathbb{R}^{n^2}$, with Euclidean space. Define a metric for $M_{nn}(\mathbb{R})$ as:

$$d: M_{nn}(\mathbb{R}) \times M_{nn}(\mathbb{R}) \rightarrow \mathbb{R}; (A, B) \mapsto \sqrt{\sum_{i,j} (A_{ij} - B_{ij})^2}$$

Since the Matrix multiplication

$$\mu: M_{nn}(\mathbb{R}) \times M_{nn}(\mathbb{R}) \rightarrow M_{nn}(\mathbb{R}); (A, B) \mapsto \left(\sum_{k=1}^n A_{ik} \cdot B_{kj} \right)_{i,j}$$

is continuous, because it is entry-wise continuous.

Since $GL_n(\mathbb{R})$ is closed under the multiplication, it has also continuous operation.

And, $GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R}); A \mapsto A^{-1}$

is also continuous, because $A^{-1} = (\det A)^{-1} \cdot \text{adj } A$

$$\text{where } (\text{adj } A)_{i,j} = (-1)^{i+j} \cdot \det A(\hat{j} \hat{i}).$$

is also multivariables polynomial, entrywise.

Consider $O_n(\mathbb{R}) = \{A \in M_{nn}(\mathbb{R}) \mid A^t = -A\}$, $SO_n(\mathbb{R}) = \{A \in O_n(\mathbb{R}) \mid \det A = 1\}$.

Then, O_n and SO_n are Compact, and SO_n is connected.

Proof. $A \in O_n(\mathbb{R}) \Leftrightarrow A^t A = I \Leftrightarrow \sum_{k=1}^n A_{ik} \cdot A_{jk} = \delta_{ij}$.

Define a map $f_{i,j}: M_{nn}(\mathbb{R}) \rightarrow \mathbb{R}; A \mapsto \sum_{k=1}^n A_{ik} \cdot A_{jk}$.

Then, it is continuous therefore

$$O_n(\mathbb{R}) = \bigcap_{\substack{1 \leq i, j \leq n \\ i \neq j}} f_{i,j}^{-1}[\{0\}] \cap \bigcap_{1 \leq i \leq n} f_{i,i}^{-1}[\{1\}] \text{ is finite intersection of closed sets,}$$

thus closed. Moreover, $\dim O_n(\mathbb{R}) \leq n$, so it is bounded. Therefore Compact.

And, $\det: O_n(\mathbb{R}) \rightarrow \mathbb{R}; A \mapsto \det A$ is continuous, so,

$SO_n(\mathbb{R}) = \det^{-1}[\{1\}]$ also closed, and bounded by subset of a bounded set.